

How much does it cost to power AI data centers?

Posted by u/mike0connor • 0 points • 16 comments

Recently, I've been seeing and reading a lot about AI data centers almost literally running out of energy, because they already put so much strain on the power system and will only put more as they grow and multiply, to the point where the companies and investors have started finding/building their own power sources. How much does AI actually const to power in actual numbers? Preferably watts and dollars.

Comments

u/[deleted] • 1 points • 2026-03-24 10:23 UTC

[removed]

u/Intelligent_Molecule • 1 points • 2025-10-07 10:45 UTC

Powering AI data centers consumes a tremendous amount of energy and can cost millions of dollars annually. A single high-end AI GPU server, equipped with multiple GPUs, typically uses between 10 to 15 kW under full load. When you scale this up, a mid-sized AI data center with around 100 such servers could demand about 1.5 megawatts (MW) of continuous power. At an average industrial electricity rate of roughly \$0.10 per kWh, this translates to about \$150 per hour, \$3,600 per day, or roughly \$1.3 million annually just in energy costs. Larger hyperscale AI data centers that draw 50 MW or more can see electricity expenses exceeding \$43 million per year. These massive energy demands often strain local power grids, prompting many AI companies and investors to invest in renewable energy projects or build their own power infrastructure to ensure reliable and cost-effective power supply.

u/ingeniousbuildIO • 1 points • 2025-10-02 15:04 UTC

depends on soo many factors

u/CryptoThousandAir • 0 points • 2025-09-30 02:18 UTC

Regardless of the type of DC, the cost could be about 9-12 cents per Kw hour. There isn't enough power from the grid, so many are relying on on-site power generation. I personally sell on site power to DC's

u/robbieboy95 • 1 points • 2025-10-05 08:28 UTC

does it sound realistic for you, to close a PPA at 15-18 [cents/kWh] for carbon-free 24/7-available energy?

u/Watt-Bitt • 1 points • 2025-09-30 00:09 UTC

It really depends on the setup... things like whether the site is under a physical PPA or vPPA, local utility tariffs, time-of-use rates, PUE/cooling design, load profile (24/7 vs variable), hardware efficiency, and even grid upgrade costs all swing the numbers. A hyperscale AI data center can draw anywhere from 10 MW to 200+ MW continuously, which at ~\$50-\$120/MWh (typical PPA ranges) works out to roughly \$20M-\$60M/yr for a 50 MW site and tens of millions more for 100+ MW sites. In other words, there's no single answer, just a wide range driven by contracts and configuration.

u/Watt-Bitt • 1 points • 2025-09-30 00:10 UTC

I have a model for this btw.

u/l0veit0ral • 1 points • 2025-09-30 00:58 UTC

DM'd you about your model

u/ImNotADruglordISwear • 2 points • 2025-09-29 23:55 UTC

I've got a theory that there's bots posting these and negative comments on literally any post that includes "datacenter" and they all make similar points yet they're all worded different. So, in a sense, using AI to make these comments about the same thing they want to "destroy" or make negative press towards.

u/looktowardward • 1 points • 2025-09-30 05:42 UTC

But why would anyone have a bot doing this?

That being said - it is absolutely weird.

u/BeardBootsBullets • 2 points • 2025-09-29 23:12 UTC

Everyone has already said the most relevant information, but I want to add that data centers are responsible for incredible power and cooling efficiencies via their economies of scale and drive for innovation.

The worst possible scenario would be to ban data centers and force their closure. If you think that data centers use too much power and water *right now,* wait until we go back to on-prem server closets.

u/smoketezt • 2 points • 2025-09-29 23:06 UTC

These vague meandering questions all seem similar lately....

u/Corbusi • 0 points • 2025-09-29 21:42 UTC

A rack is a frame that is 600mm wide by 1200mm deep x 2100mm high (basic model) where computers are vertically stacked.

The maximum amount of power that racks used to need was 8kW.

Today we are seeing AI racks that require 120kW. That is 15 times more power per rack.

In 5 years, it is expected some racks will be 600kW. That is 75 times more power per rack.

A 20MW Data Centre with poor power use efficiency (PUE) might have 1500 racks at 8kW. If the power available cannot change, then drawing 15 times more power for a 120kW rack will result in 1/15 of the original 1500 racks so 100 racks. A lot of empty space in the Data Halls. But the same amount of heat being generated.

A computer is effectively a heater. It converts electricity to heat.

Cost of power across the world varies.

If 1 unit of electricity cost \$1 in the USA, then the same amount in the UK costs 32 cents and the same in Saudi Arabia costs 5 cents.

The yanks power network is democratically driven by commercial profits. The UK the same. The Saudis have lots of oil, which they use to generate electricity. The Government also subsidises the electricity supply to reduce costs.

u/looktoward • 2 points • 2025-09-30 05:40 UTC

\> If 1 unit of electricity cost \$1 in the USA, then the same amount in the UK costs 32 cents

not even close to true. Industrial power in the UK is about 26pence/kwh or about 35c/kwh. Industrial power in the US varies from 7c to 40c/kwh.

u/robbieboy95 • 2 points • 2025-10-05 08:26 UTC

US has cheap power in general. Europe has one of the highest energy prices worldwide

u/modaloves • 2 points • 2025-09-29 21:17 UTC

There's no real standard for what counts as an “AI datacenter” vs a conventional one. Some companies throw the term around a lot for marketing purpose, while others avoid it even when the specs look basically the same.

From what I've seen in press releases from various companies, GPU-intensive datacenters usually pull 50MW~1GW.

u/refboy4 • 24 points • 2025-09-29 21:08 UTC

Not intended to be offensive, but this kinda like asking how fast your shoes run.

It depends on the efficiency and age of the equipment, if you are running that equipment at the most efficient rate *usually 80%), where it's located (free cooling available?), how much power costs from that local utility, what if any tier system or contract with that utility has been set in place...

u/looktoward • 3 points • 2025-09-29 20:51 UTC

Data centers pay industrial power tariffs based on their size. Maybe 8c/kwh regardless of size. Lets say its a 500MW data center.

u/CryptoThousandAir • 1 points • 2025-09-30 02:16 UTC

More like 9-12c

u/looktoward • 1 points • 2025-09-30 02:23 UTC

Depends where. I got cheap power :)

u/looktoward • 1 points • 2025-09-30 02:23 UTC

Depends where. I got cheap power :)

u/CryptoThousandAir • 0 points • 2025-09-30 02:24 UTC

Very true. Some states still have cheap/clean power. The deals I work are in gigawatts so not everyone has that to give